Chapter 11 Answers

September 8, 2023

Note : Smullyan must have made a typo when defining Σ_0 formulae, as he said that the existential type is $\exists x(x \leq \overline{n} \supset F(x))$, which makes no sense, as this is a valid because there is always some x that is greater than n, thus making the statement true. It seem that what he meant to say was $\exists x(x \leq \overline{n} \wedge F(x))$, and I will think of Σ_0 formulas as such. (I later checked pp. 94 of the Begginer's Further Guide, and his typo was confirmed, as he defines the existential case with a $\wedge$, not a $\supset$).

Problem 1

(a):
x *div* y can be represented by the formula:
$\exists z(z \times x = y)$
(b):
x *pdiv* y can be represented by the formula:
x *div* $y \wedge (\sim(x = 1) \wedge \sim(x = y))$
(c):
$x < y$ can be represente by the formula:
$\exists z(\sim(z = 0) \wedge z + x = y)$
(d):
prm x can be represented by the formula:
$\sim\exists z(z$ *pdiv* $x)$

Problem 2

Consider any set or relation that is expressible by some formula the includes $\leq$. Well, we can define $x \leq y$ as: $\exists z(z + x = y)$, thus meaning the same formula is expressible in terms of plus and minus.

Problem 3

We are being asked to prove that $y > 0$ and $x = 2^{L(y)}$ are true iff the following three conditions hold:
$C_1 : x - 1 \leq y \leq 2 \times x - 1$
$C_2 : x$ is a power of 2
$C_3 : y \neq 0$

Proof. On the one hand, assume that the three conditions C_1, C_2, C_3 all hold. Well, by C_3, $y \neq 0$. Hence, because y must be a positive integer, it follows that $y > 0$. Next, we know by C_2 that x is a power of 2, and thus, $x = 2^m$ for some m. Hence, substituting into C_1, we get:
$2^m - 1 \leq y \leq 2 \times 2^m - 1$
Then, by $(*)$, we know that $m = r$ for $r = L(y)$. Hence, we know that $x = 2^m = 2^r = 2^{L(y)}$, thus completing this case of the proof.

On the other hand, assume that $y > 0$ and $x = 2^{L(y)}$. Hence, C_3 holds, as $y \neq 0$. Also, because $x = 2^{L(y)}$, C_2 holds, as x is a power of 2. Lastly, because $x = 2^{L(y)}$ where $L(y) = r$ for r the length of y, we know $x = 2^r$, and thus we can substitute, yielding $x - 1 \leq y \leq 2 \times (x - 1)$ by $(*)$. □

Problem 4

C_1 is true iff:
$(\exists z \leq x)(z = x + 1 \wedge z \leq y \wedge y \leq 2 \times z)$

Problem 5

(a):
$Pow_2(x)$ can be expressed by the formula:
$(\forall y \leq x)((y \; div \; x \wedge y \neq 1) \supset 2 \; div \; x)$
(b):
First, $x \neq y$ can be expressed by the formula $\sim(x = y)$. So, we can represent the desired relation with the formula:
$x \neq 0 \wedge y \neq 0 \wedge (\exists v \leq z)(v = 2^{L(y)} \wedge z = x \times v + y)$
by Problem 3. We can tell this is Σ_0, as, because all of C_1, C_2, C_3 are Σ_0, we have their conjunction $v = 2^{L(y)}$ also being Σ_0.

Problem 6

(a):
Well, because the concatenation operation is assosciative, $x * y * z = (x * y) * z$. Thus, by Problem 5(b), $(x*y)*z$ is Σ_0. Hence, the relation under consideration can be represented by the Σ_0 formula:
$0 \neq x_1 \wedge 0 \neq x_2 \wedge 0 \neq x_3 \wedge (\exists v \leq z)(v = x_1 * x_2 \wedge z = v * x_3)$
(b):
The base case was proven in Problem 5(a). Now, make an induction assumption. Hence:
$0 \neq x_1 \wedge ... \wedge 0 \neq x_n \wedge x_1 * x_2 * ... * x_n = y$ is Σ_0. In the case for $n+1$, we have the following Σ_0 formula representing the desired relation by the induction assumption, as well as Problem 5(a): $0 \neq x_1 \wedge ... \wedge 0 \neq x_n + 1 \wedge (\exists z \leq y)(z = x_1 * x_2 * ... * x_n \wedge y = z * x_n + 1)$

Problem 7

(a):
$x\ B\ y$ can be represented by the Σ_0 formula:
$(\exists z \leq y)(xz = y \vee x = y) \wedge (x \neq 0 \wedge y \neq 0)$ (b):
$x\ E\ y$ can be represented by the Σ_0 formula:
$(\exists z \leq y)(zx = y \vee x = y) \wedge (x \neq 0 \wedge y \neq 0)$ (c):
$x\ P\ y$ can be represented by the Σ_0 formula:
$(x\ B\ y \vee x\ E\ y) \vee (\exists z_1 \leq y)(\exists z_2 \leq y)(z_1 x z_2 = y)$

Exercise

We are considering some number z satisfying the following two conditions:
(1): z contains at least one frame.
(2): If f is the largerst frame in z, then there is at least one string in z of the form $ffafbff$ where f is not a part of a or b.
We are to show that this number z encodes one and only one sequence of positive integers.

Proof. Let us recall the definition of what it means for a finite sequence S of ordered pairs of positive integers to be encoded in some z (on pp. 205):
C_1: The longest frame f in z contains a string of 1's longer than any string of 1's in any of the numbers in S.
C_2: Every ordered pair (a, b) in S is found within z in a string of the form $ffafbff$, where f occurs in neither a nor b, and with f being the longest frame

occuring in z.
C_3: The pairs $(a_1, b_1), ..., (a_n, b_n)$ are found encoded in the sequence as we move form left to right in the dyadic notation for z. Well, Smullyan either made a mistake in his definitions, or I misunderstood his definitions, as the following example disproves the statement. Consider the number z in dyadic notation: 21221211212111212212 with 212 being the longest frame in z. The only possible finite sequence of integers that z could encode would be the a and b which are listed in z, namely $11, 111$. Well, by condition C_1, for z to be a code number for this sequence, the longest frame in z must have a string of ones longer than the longest string of ones in the integer sequence. However, 212 has only one 1, while $11, 111$ both have longer strings, thus seemingly disproving the statement. $\square$

Problem 8

We are attempting to find some Σ_0 relation $K_0(x, y, z)$ such that for any finite sequence of ordered pairs of natural numbers, there is a number z such that for all numbers x and y, the relation $K_0(x, y, z)$ holds iff (x, y) is one of the ordered pairs in S. Let us define the set S_1 to be the set of all ordered pairs of integers $(x + 1, y + 1)$ such that the ordered pair (x, y) is a member of S. So, we then define $K_0(x, y, z)$ to be the following Σ_0 relation:
(1) $(\exists n \leq z)(\exists m \leq z)(n = x + 1 \wedge m = y + 1 \wedge K_1(m, n, z))$
And we know that there is some $K_1(m, n, z)$ relation such that z is a code number for S' and $K_1(m, n, z)$ holds iff $(m, n) \in S$. Hence, (1) holds iff
$K_1(x + 1, y + 1, z)$ (for the z that makes $K_1(m, n, z)$ encode S) holds iff $(x + 1, y + 1) \in S_1$ iff $(x, y) \in S$, thus completing the proof.

Problem 9

Proof. We are to show that the realation $g(\overline{x}) = y$ is Σ_1, that is, there is some formula of the form $\exists z R(x_1, ..., x_n, z)$ that represents $g(\overline{x}) = y$ where the relation $R(x_1, ..., x_n, z)$ is Σ_0. Well, let us first consider the set S of a sequence of ordered pairs of natural numbers:
$(0, g(\overline{0})), (1, g(\overline{1})), ..., (n, g(\overline{n}))$
where $n = x$ in the equation $g(\overline{x}) = y$. To be clear, every $g(\overline{i})$ in S shall not be the Dyadic Gödelian representation that $g(\overline{i})$ is, but instead is the *decimal number* that is designated by the *Dyadic Godelian representation for* i. Thus, every member of an ordered pair of S is a natural number. Now, let us recall the fact that, for each positive integer $i \leq x$, $g(\overline{i+1}) = g(\overline{i}) * 122$, and that $i = 0 \supset g(\overline{i}) = 12$. Now, what we wish to show next is that, if $g(\overline{x}) = y$, then there is some sequence of natural numbers such that:
(1) (m, n) is a member of S

(2) If some (a, b) is a member of S, then so is (c, d) for $(a, b) = (c + 1, d * 122)$ or $(a = 0, b = \overline{4})$.
Well, by definition, every $g(\overline{x}) = y$ means that either $x = 0,\ y = \overline{4}$, or that, for some number a, b, $(x, y) = (a + 1, b * 122)$, and similarly for (a, b), and so on. Thus S contains this sequence. It can be seen by induction on every x that $g(\overline{x}) = y$ holds for every (x, y) in S, and thus, if we have some sequence S according to these conditions, it follows that $g(\overline{m}) = n$ holds. Hence, we have the following Σ_1 formula expressing $g(\overline{x}) = y$:
$\exists z[K_0(x, y, z) \wedge (\forall a \leq z)(\forall b \leq z)(K_0(a, b, z) \supset$
$[(a = 0 \wedge b = \overline{4})$
$\vee\ (\exists c \leq z)(\exists d \leq z)(K_0(c, d, z) \wedge a = c + 1 \wedge b = d * 122)])]$
Now let's be clear as to why this formula works. Assume that the above formula holds. Thus there is some z such that z is a code number for at least $(x + 1, y + 1)$, where the set S_1 it is a code number for has the corresponding set S (defined in Problem 8) which clearly satisfies conditions 1, 2, and thus $g(\overline{x}) = y$. On the other hand, assume that $g(\overline{x}) = y$. Thus, there is *some* number z such that z enocdes the set S_1 which corresponds to the set S which is one of the sets satisfying conditions 1, 2 for (x, y), thus meaning the formula holds. *Note*: We very well could have instead written $K_0(x, y, z)$ as $(\exists n \leq z)(\exists m \leq z)(n = x + 1 \wedge m = y + 1 \wedge K_1(m, n, z))$, but we wrote it as we did for simplicity. □

Exercise 2

We are to show that the relation $x^y = z$ is Σ_1.

Proof. Consider a set S of a sequence of ordered pairs such that:
(1): $(y, z) \in S$
(2): If $(a, b) \in S$ than either $a = 0 \wedge b = 1$ or there is some $(c, d) \in S$ such that $(a, b) = (c + 1, d \times x)$
where x, y, z are defined from the equation $x^y = z$. On the one hand, assume there is a set S satisfying conditions (1) and (2). It can easily be shown with generalized induction, where the component of (a, b) is $(a - 1, b/x)$ such that $(0, 1)$ has no components, that, for every $(a, b) \in S$, $x^a = b$, and thus, in particular, $x^y = z$. Conversley, assume that $x^y = z$. Hence we place $(y, z) \in S$, and then it is obvious that we can satisfy condition (2), as, by definition of a power, either $y = 0 \wedge z = 1$ or we can place the ordered pair (a, b) such that $(y, z) = (a + 1, b \times x)$ (we can do this $y - 1$ times, as a power x^y is defined to have y factors of x). Thus $x^y = z$ iff there is a set S satisfying conditions $(1), (2)$. Hence, using Lemma K_0, we receive the following formula Σ_1 formula:
$\exists z[K_0(c, d, z) \wedge (\forall a \leq z)(\forall b \leq z)(K_0(a, b, z) \supset$
$[(a = 0 \wedge b = 1) \vee (\exists m \leq z)(\exists n \leq z)(a = m + 1 \wedge b = d \times e)])]$
that expresses the relation $x^y = z$. For, assume the formula holds. Hence, there

is some z such that z is a code number for a sequence of natural numbers in a set S such that the set S_1 obviously satisfies conditions $(1), (2)$. Conversley, if $x^y = z$, then there is some sequence of pairs of natural numbers satisfying conditions $(1), (2)$, and the corresponding set S_1 can be encoded by *some* code number z. $\square$

Problem 10

We consider some Σ_0 relation $R(x, y, z_1, ..., z_n)$ and consider the relation $\exists x \exists y R(x, y, z_1, ..., z_n)$. We are to show that this is Σ_1. Well, the above is equivalent to the following Σ_1 relation:
$\exists a[(\exists x \leq a)(\exists y \leq a)R(x, y, z_1, ..., z_n)]$
For, assume the above formula holds. This obviously implies that the initial relation holds. Conversley, assume that the initial formula holds. Then, there is some a such that the above Σ_1 relation holds, namely some a such that $a \geq x \wedge a \geq y$. Then, assume the relation $R(x, z_1, ..., z_n)$ is Σ_1. By definition, it follows that it is equivalent to some relation $\exists y R_1(x, x_1, ..., x_n, y)$ where $R_1(x, x_1, ..., x_n, y)$ is Σ_0. Hence, by the above result, we have $\exists x R(x, z_1, ..., z_n)$ being equivalent to the Σ_1 relation $\exists x \exists y R_1(x, x_1, ..., x_n, y)$. (It should be noted that, for both of these problems, it is of course not a requirenment that the variables being quantified are the first and second. The proofs that these two relation are Σ_1 works for any free variable at all in the relation are similar , just by replacing x by x_i, and y by y_i, and so forth).

Problem 11

We are attempting to show that there is some Σ_1 function $r(x, y)$ such that for any e the Gödel number of any expression E, and any number n, $r(e, n)$ is the Gödel number for $E[\overline{n}]$. Well, $E[\overline{n}]$ is:
$\exists v_1(v_1 = \overline{n} \wedge F(\overline{n}))$
Well, we let $\exists v_1(v_1 =$ have the Gödel number g, $\wedge$ is g_8 while $F(\overline{n})$ has the Gödel number f, and) has the Gödel number g_4. So, the Gödel number of the above formula is $g * g(\overline{n}) * g_8 * f * g_4$. So, $r(x, y)$ is this formula. Now, we must show that $r(x, y) = z$ is Σ_1. Well, this is equivalent to:
$\exists w(w = g(\overline{n}) \wedge g * w * g_8 * f * g_4 = z)$
Recall that $w = g(\overline{n})$ was shown to be equivalent to some formula of the form $\exists z R(x, y, z)$ where $R(x, y, z)$ is Σ_0. Hence, we can rewrite the above formula as:
$\exists w \exists v(R(w, y, v) \wedge g * w * g_8 * f * g_4 = z)$
Then, because the realation $R(w, y, v) \wedge g * w * g_8 * f * g_4 = z$ is Σ_0, the above formula is Σ_1 by Problem 10.

Note to self : To be *very* clear (as I spent numerous hours highly puzzled over this), when Smullyan says that we have shown $r(x, y) = z$ to be arithmetic, he does *not* mean $r(x, y)$ to be any relation at all (as I wrongly thought); he of course means it to be the $r(x, y)$ defined in this problem.

Problem 12

Assume A is Σ_1. Hence there is a Σ_0 relation $R(v, w)$ such that $v \in A$ iff $\exists w R(v, w)$. Also, $r(x, x) = y$ has been shown to be Σ_1, and hence there is a Σ_0 relation $R(x, y, z)$ such that $r(x, x) = y$ holds iff $\exists z R(x, y, z)$ is true. Hence $r(x, x) = y \wedge y \in A$ iff $\exists z \exists w(R_1(x, y, z) \wedge R_2(y, w))$, and this is Σ_1 by Problem 10, thus meaning $r(x, x) = y \wedge y \in A$ is Σ_1, and hence:
$\exists y(r(x, x) = y \wedge y \in A)$, that is, the formula that expresses $A^{\#}$, is also Σ_1 (also by Problem 10).

Problem 13

We consider two Σ_1 sets A, B. Thus, A, B are expressed by the two formulas $\exists z R_1(x, z)$, $\exists y R_2(x, y)$, respectively. Hence, the Σ_1 formula $\exists z(\exists x \leq z)(\exists y \leq z)(R_1(v, x) \wedge R_2(v, y))$ clearly represents $A \bigcap B$, as it is true iff $v \in A \wedge v \in B$. Also, the Σ_1 formula $\exists z(\exists x \leq z)(\exists y \leq z)(R_1(v, x) \wedge R_2(v, y))$ represents $A \bigcup B$, as it is true iff $v \in A \vee v \in B$.

Problem 14

We have already shown that every arithmetic set has a Gödel sentence, so the below result holds true:
Well, we consider any set W_0 of Gödel numbers of true sentences that is arithmetic. Thus the complement $\widetilde{W_0}$ of W_0 is also arithmetic, and therefore has some Gödel sentence X. Hence, by definition of a Gödel sentence, X is true iff x^*, for x^* the Gödel number of X, is in $\widetilde{W_0}$ iff $x^* \notin W_0$. Hence, X must be true, as if it was not, $x^* \in W_0$ and X would be false, contrary to the assumption that only Gödel numbers of true sentences were members of W_0.